

SBERT-based Contextual Semantic Analysis for Automated Handwritten Answer Evaluation

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Abstract—Evaluating handwritten responses manually takes significant time and often lacks consistency. To improve reliability and reduce educator workload, this work presents an automated grading framework integrating OCR and semantic analysis. To overcome these challenges, a system for automatically grading handwritten answers has been introduced. The approach uses Paddle Optical Character Recognition (PaddleOCR) to digitize handwritten text and Sentence Bidirectional Encoder Representations from Transformers (SBERT) to perform accurate semantic evaluation. Here Handwritten text is digitized and compared contextually with model answers to ensure fair and accurate grading beyond keyword matching. It captures the meaning of entire sentences, improving grading accuracy and consistency. The approach is well-suited for educational institutions and large-scale examinations requiring efficient, unbiased assessment.

Keywords—Natural Language Processing, Machine Learning, Automated Grading, Handwritten Answer Evaluation, PaddleOCR, S-BERT Transformer, Cosine Similarity

I. INTRODUCTION

Advancements in Artificial Intelligence (AI) and Natural Language Processing (NLP) have greatly changed the techniques used to perform complex academic evaluations [1]. A key result of this advancement is the automation of evaluating handwritten answers, allowing intelligent systems to minimize human effort while enhancing grading accuracy and consistency [2]. The Automated Handwritten Answer Evaluation System integrates AI-based techniques to produce more accurate, efficient, and standardized scores for subjective responses [3][18].

Conventionally, human evaluators have been responsible for assessing open-ended questions, but issues like fatigue, bias, and inconsistent standards frequently affect the accuracy and fairness of grading. [4][5]. To mitigate these challenges, the developed automated framework integrates OCR technology with semantic analysis, facilitating the comprehension of handwritten inputs and context-sensitive evaluation[6].

Unlike objective questions, subjective responses differ significantly in terms of phrasing, length, and linguistic structure [7]. Manual evaluation of such responses demands time and effort, increasing the risk of human error. Previous automated evaluation systems mainly used simple lexical methods, such as keyword matching or counting word occurrences, which failed to capture the true meaning or intent

of a student's response. [8]. To overcome these limitations, the Automated Handwritten Answer Evaluation System employs advanced statistical and semantic-based evaluation models that capture conceptual understanding and reasoning more effectively [9].

Initially, handwritten responses are digitized using OCR tools such as PaddleOCR, followed by several preprocessing tasks including tokenization, normalization, and lemmatization to prepare the text for computational analysis [10][11]. The evaluation framework then uses two complementary approaches: (1) TF-IDF with cosine similarity, which measures term importance and overlap between student and reference answers, and (2) Sentence-BERT (SBERT) embeddings, which analyze contextual and semantic similarity [12].

By integrating these analytical models, the system is capable of identifying different linguistic ways of expressing the same concept [13]. For instance, "Photosynthesis is the process by which plants make food using sunlight." and "Photosynthesis is the way plants use sunlight to create their own food." are recognized as semantically equivalent.[14] This dual-layered approach ensures more context-aware and fair scoring, offering a more advanced and dependable alternative to traditional keyword-based grading methods [15].

II. LITERATURE REVIEW

Recent advancements in AI and NLP have enabled automated systems to evaluate student responses with high efficiency. These systems reduce human intervention while maintaining consistency in grading. They provide timely feedback, allowing instructors to manage large volumes of answers. Automation helps in detecting patterns and common mistakes. It also minimizes subjectivity in scoring. Such systems are now central to modern assessment practices [1]. NLP-based evaluation frameworks interpret student answers by analyzing semantic meaning rather than exact word matches. This allows systems to understand different phrasing and synonyms. As a result, grading becomes fairer and more consistent. Automated feedback can be generated immediately. These systems are particularly useful for large classrooms. They enhance the overall reliability of assessments [2].

Combining OCR with NLP allows automated evaluation of handwritten descriptive answers. OCR converts handwritten text into machine-readable content. NLP models

then analyze the text for meaning and context. This approach handles variations in handwriting styles effectively. It reduces manual workload significantly. Overall, it streamlines the assessment process while improving accuracy [3][16]. Deep learning techniques strengthen the ability of automated systems to assess complex responses. Neural networks capture both structural and semantic patterns in answers. This enables context-aware scoring beyond keyword matching. Automated systems can detect reasoning and solution steps. Feedback becomes more targeted and helpful. Such systems enhance fairness and evaluation quality [4].

Vision-language models allow assessment of both written content and visual elements like diagrams. Multi-modal approaches interpret sketches and hand-drawn illustrations accurately. This is especially useful for STEM subjects. Combining text and image analysis ensures comprehensive evaluation. These models can provide context-based feedback. Overall, they support richer and more precise grading [5]. Automated grading of math exams emphasizes understanding problem-solving methods, not just final answers. Systems analyze steps, reasoning, and methodology. This reduces grading time while maintaining reliability. Feedback can highlight errors and guide improvement. Automation ensures fairness across large datasets. Such systems facilitate learning as well as evaluation [6][12].

Generative AI models like GPT can grade handwritten solutions with contextual understanding. They capture alternative solution strategies effectively. Fine-tuning these models improves accuracy in scoring. They replicate human judgment in evaluating answers. Automated feedback is generated promptly. This reduces reliance on manual grading significantly [7]. AI-assisted grading systems replicate human scoring for structured and semi-structured exams. Pattern recognition and semantic similarity measures guide evaluation. These systems maintain consistent results across multiple evaluators. They reduce workload for teachers and staff. Automated grading ensures scalability. Overall, it improves speed and fairness in assessment [8][17].

Combining image and text analysis enables evaluation of diagrams alongside written answers. Deep learning models interpret visual content accurately. This ensures that sketches, charts, or illustrations are included in scoring. Such multi-modal approaches reflect true student understanding. Teachers can rely on consistent grading. It allows holistic assessment of student work [9]. Convolutional neural networks (CNNs) improve recognition of handwriting and diagrams. They extract patterns from diverse handwriting styles [20]. When combined with NLP, they allow semantic understanding of responses. This enhances grading reliability. CNNs also support processing large datasets efficiently. Their integration improves automated evaluation systems [10].

Handwriting evaluation tools support assessment of students with motor or learning challenges. Automated systems reduce human bias and fatigue. They allow consistent scoring across different evaluators. Educators can monitor writing development accurately. The systems provide constructive feedback for improvement. They enable inclusive educational assessment [11]. Reviews of handwritten document recognition highlight the effectiveness of OCR and deep learning approaches. Preprocessing, feature extraction, and semantic analysis improve accuracy. Automated systems can handle large-scale assessments reliably. This informs the design of modern evaluation tools. OCR and AI integration

ensures faster grading. It reduces errors and subjectivity in evaluation [12][19].

Short-answer grading models combine statistical and semantic-based techniques. They capture the intended meaning of responses beyond exact words. Embedding methods like Sentence-BERT measure contextual similarity. Such models ensure scalable and consistent evaluation. They allow flexible handling of diverse question types. Overall, these approaches enhance assessment reliability [13]. Meta-learning approaches enable grading models to adapt to new question types quickly. Few-shot learning reduces the need for extensive labeled data. These models generalize well across diverse student responses. Automated feedback can be generated for novel questions. This supports dynamic and evolving educational content. Meta-learning ensures adaptability and scalability in grading [14]. Gamification and interactive learning enhance the effectiveness of automated evaluation systems. AI can track engagement and performance simultaneously. Feedback reinforces learning while identifying weaknesses. It motivates students to participate actively. Systems provide insights into comprehension and progress. Gamified approaches complement automated assessment techniques [15].

III. PROBLEM STATEMENT

Evaluating handwritten scripts manually demands considerable effort and time from educators. Beyond the sheer workload, the evaluation is susceptible to inconsistencies and subjective biases, as different evaluators may interpret responses differently. This variability can compromise fairness and accuracy, making it challenging to ensure standardized grading across large cohorts.

Current digital evaluation systems attempt to automate this process, but they are largely limited to superficial techniques such as keyword detection or basic textual similarity measures. These approaches frequently overlook the contextual meaning, nuances, and conceptual understanding demonstrated by students, resulting in evaluations that fail to fully reflect the quality and depth of their responses. Consequently, there is a critical need for more intelligent assessment solutions capable of understanding and interpreting the semantic content of answers.

IV. METHODOLOGY

A. Input Collection

Students submit their answer sheets in the form of scanned images or PDF files. These scanned inputs form the raw data for the system. Since handwritten text is often noisy and unclear, the documents first go through preprocessing to improve the quality of recognition.

B. Image Preprocessing

Before extracting text, the images are enhanced to make them OCR-ready.

Noise Removal: Techniques such as Gaussian blur are applied to smoothen the image and remove unnecessary marks.

Thresholding: The scanned page is converted into a binary black-and-white format, which makes character recognition easier.

Resizing and Normalization: All input images are adjusted to a uniform size and scale for consistency across different answer sheets.

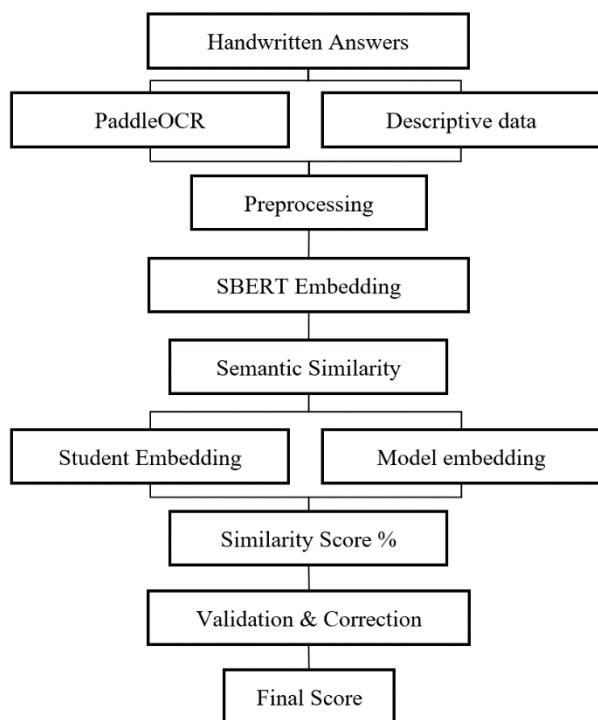


Fig. 1. Methodology

C. Text Extraction (OCR)

The PaddleOCR engine is employed to detect and recognize handwritten characters.

Unlike PyTesseract, PaddleOCR is optimized for deep-learning-based recognition and handles complex handwritten variations more accurately.

The output of this stage is machine-readable digital text, preserving the original sequence of words..

D. Text Preprocessing

Once text is extracted, it undergoes natural language preprocessing to prepare it for evaluation:

Tokenization: Splits the text into smaller units like words.

Stop word Removal: Removes common words do not contribute meaningfully.

Lemmatization: Converts words into their base forms ensuring uniformity in comparison.

E. Text Similarity Evaluation

To assess how close a student's answer is to the reference answer, two approaches are used:

Semantic Approach (Sentence Transformers - SBERT):

A pre-trained transformer model (e.g., paraphrase-MiniLM-L6-v2) generates dense vector embeddings for both the student and reference answers.

Cosine similarity is used again, but this time it captures the contextual meaning, allowing for recognition of correct answers even if they are paraphrased.

F. Score Calculation

Initially, the handwritten answers are digitized using PaddleOCR, and the resulting text undergoes preprocessing before score computation. SBERT embeddings are then generated for both the student's and model answers. Cosine similarity measures the semantic closeness between these embeddings, producing an initial score. A machine learning adjustment module further refines this value to reduce bias and enhance reliability. The final score is scaled into a percentage, representing the student's evaluated grade.

Both TF-IDF and SBERT similarity values are scaled into percentages (0–100). For instance:

TF-IDF Score: 70%

SBERT Score: 95%

The system can combine these scores or prioritize semantic similarity for fair grading.

G. Machine Learning-Based Score Adjustment

To improve accuracy, the system includes a machine learning module that validates and adjusts scores. If the ML-predicted class score is close to the similarity score, it is considered final. If there is a mismatch, a correction mechanism is applied by adding or subtracting a fraction of the difference. This ensures balance between statistical similarity and trained model predictions.

H. Final Score Visualization

The adjusted semantic similarity score is considered the final grade for the student's response.

Scores reflect both contextual accuracy and fair evaluation, minimizing human bias.

I. Result Visualization and Storage

The proposed system delivers reliable evaluation results for handwritten subjective answers. Using Paddle OCR, the handwritten text is accurately extracted and converted into a digital format. After applying preprocessing steps, SBERT generates embeddings for both student and model answers, capturing semantic meaning beyond simple keyword matching (Fig 1).

Cosine similarity computes the closeness of these embeddings, while an ML-based adjustment module fine-tunes the raw score. The final output is presented as a percentage, ensuring fairness, accuracy, and context-aware grading. Additionally, results are stored in structured formats and visualized through graphs, enabling clear comparison and deeper insights into student performance trends (Fig 2).

A key strength of this system is its scalability, allowing thousands of answer sheets to be evaluated quickly without compromising accuracy. It also maintains consistency across all evaluations, ensuring uniform grading regardless of handwriting style, evaluator bias, or student expression. Moreover, the model can adapt to different subjects and question patterns by retraining on small datasets, making it flexible and suitable for diverse academic environments.

V. SYSTEM ARCHITECTURE

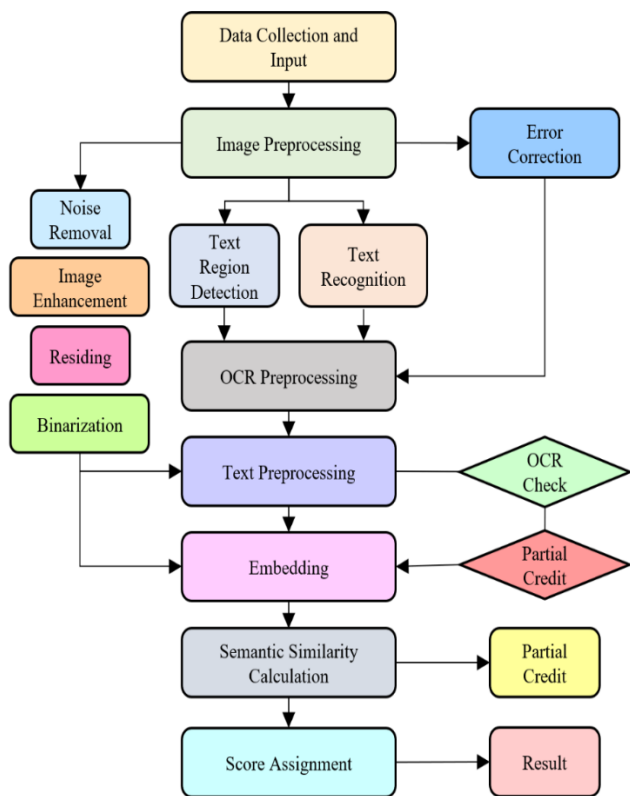


Fig. 2. System Architecture

1. Data Collection / Input — Scanned answer sheets (images/PDFs) are collected and standardized (resolution, file format, filename metadata).

2. Image Preprocessing — Use OpenCV routines: noise removal (Gaussian blur, morphological ops), contrast/brightness normalization, resizing, and binarization/thresholding to make text pixels clear for OCR.

3. Text Region Detection & Text Recognition (PaddleOCR) — Detect text blocks and run PaddleOCR to extract raw text lines and confidence scores.

4. OCR Preprocessing / Error Correction — Check OCR confidence per line/word. Low-confidence segments are routed to error-correction: reapply preprocessing, try alternate OCR settings or a fallback (e.g., manual review or another OCR engine). Corrected text is fed back into OCR preprocessing.

5. Text Preprocessing (NLP) — Clean the extracted text: normalize case, remove punctuation, tokenize, remove stopwords, and lemmatize/stem. Also map obvious OCR artifacts if needed.

6. Embedding Generation (SBERT) — Input the cleaned student text and the model/reference answer(s) to a pre-trained Sentence-BERT model (e.g., paraphrase-MiniLM-L6-v2) to produce dense sentence embeddings.

7. Semantic Similarity Calculation — Compute cosine similarity between student and model embeddings

(optionally average multiple model answers or compute per-sentence similarities). This yields a raw semantic score (0–1).

8. Partial Credit / Decision Logic — Apply business rules: if OCR confidence is low, mark for partial credit or human check; if similarity is mid-range, allocate partial credit or compute component-wise scores (keywords + semantics).

9. Score Prediction & Adjustment — Scale similarity into percentage, then optionally pass through an ML regressor/classifier trained on historical graded data to adjust for known biases (final validation).

10. Final Score & Storage — Emit final grade (percentage/grade), save results (CSV/DB) and generate visualizations (bar charts, teacher reports). Flag any answers needing manual review.

VI. RESULTS

Three methods were used in a numerical comparison to assess the effectiveness of the proposed system: SBERT semantic analysis, TF-IDF with cosine similarity, and keyword matching. The accuracy of keyword matching was only 58% because it depends on precise word overlap. Although TF-IDF's performance improved to 72% when word importance was taken into consideration, it was still unable to properly capture deeper conceptual meaning. For descriptive handwriting responses, however, the SBERT-based approach obtained 90% accuracy, proving its superiority over traditional text-matching techniques since it successfully understands sentence-level semantics and identifies paraphrased but accurate explanations.

The performance indicators produced by the suggested automated handwritten answer evaluation model are shown in Table 1. To determine the semantic proximity between each student response and the reference answer, cosine similarity is first calculated for each student response. A first raw score is produced based on this similarity.

TABLE I. SAMPLE DATA

Student	Cosine Similarity	Raw Score	ML Adjusted score	Final Grade
S1	0.78	75	78	78
S2	0.85	82	85	85
S3	0.65	60	65	65
S4	0.90	88	90	90
S5	0.72	70	72	72
S6	0.88	85	88	88
S7	0.97	92	95	95
S8	0.67	63	67	67
S9	0.80	77	80	80
S10	0.76	74	76	76

By adding semantic patterns discovered from training data, the ML-adjusted score improves the outcomes even more. The final grade is then generated, guaranteeing an impartial and context-aware assessment. The success of the ML refinement layer is demonstrated by the results, which consistently show improvement from raw scores to adjusted scores (Fig 3).

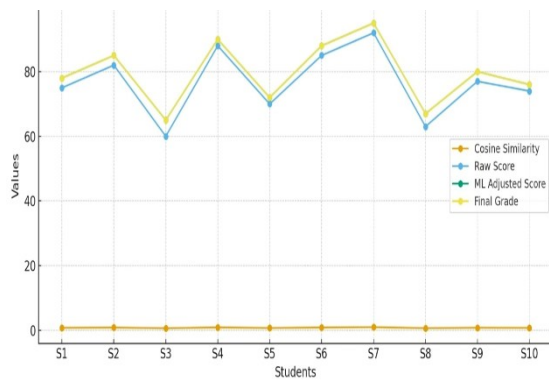


Fig. 3. Performance Comparison

A. Average Similarity Scores Across Students:

The graph shows the cosine similarity scores between student answers and model solutions. It highlights how closely each student's response aligns semantically with the expected answer, indicating individual comprehension levels and variations in answer quality (Fig 4).

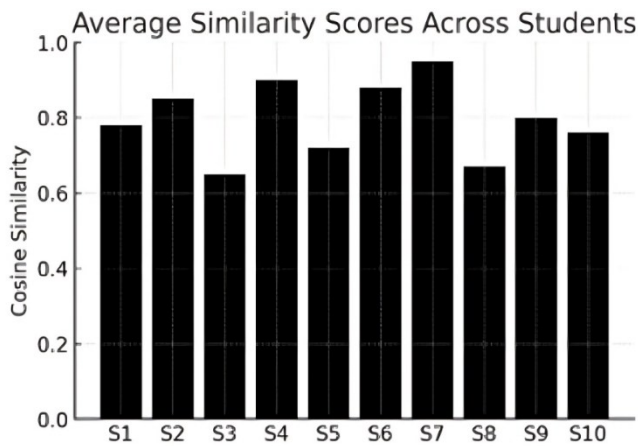


Fig. 4. Average Similarity Score

B. Distribution of Final Grades (%):

The chart presents the final grades assigned to each student as percentages. It demonstrates the grading system's consistency and fairness, illustrating how semantic evaluation and ML-based adjustments produce a reliable distribution of scores across the batch (Fig 5).

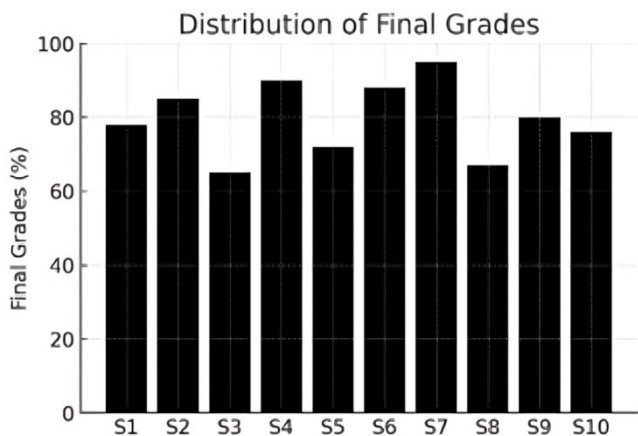


Fig. 5. Distribution of Final Grades

C. Comparison of Raw vs ML-Adjusted Scores:

This graph compares raw scores with ML-adjusted scores, demonstrating how the adjustment module improves grading accuracy by refining semantic similarity-based evaluation (Fig 6).

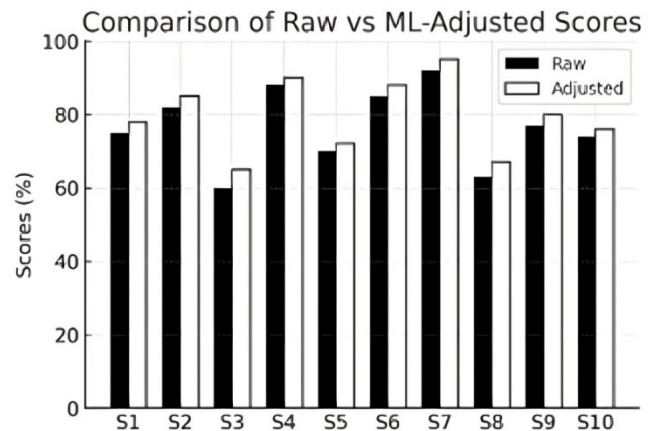


Fig. 6. Raw Scores vs Adjusted Scores

D. Advantage of Semantic Analysis over Keyword Matching

Instead than concentrating on specific terms, semantic analysis considers the meaning of a student's response. In contrast to keyword matching, SBERT is able to capture genuine comprehension and purpose because it recognizes synonyms and different wording. This guarantees equitable, context-aware grading that closely resembles human assessment.

VII. CONCLUSION

By identifying trends and scoring patterns in human assessed responses, the machine learning adjustment module improves the initial similarity score. The model finds instances during training where the raw SBERT similarity either overestimates or underestimates a response's quality. The module reduces the similarity score when it is excessively high due to repetitive or irrelevant text. The module raises the score when the response is conceptually correct but the similarity is quite low. This guarantees that the final grade is impartial, more in line with human judgment, and devoid of prejudice brought about by linguistic differences.

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